Same Base Rate, Different Skill: Why Precision at a Fixed Recall Does Not Rank Cross-Region Harmful-Algal-Bloom Forecasting Skill

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Abstract. An operational harmful-algal-bloom (HAB) early-warning system is conventionally summarised by precision at a fixed, policy-relevant recall, and that number is compared across regions as if it measured the same thing everywhere. We report a controlled demonstration, from 5 regional HAB early-warning models built on one shared pipeline, that it usually does not: precision at a fixed recall is dominated by each region's own bloom base rate, not by how much the model has actually learned about that region's blooms. The central evidence is a matched-prevalence pair: the Arabian Sea and Bay of Bengal sit within half a percentage point of each other in base rate at every horizon tested and report nearly identical precision at the longest horizon (0.202 against 0.187), yet differ by 2.8× in skill lift over a persistence baseline at that same horizon — the region reporting the *lower* precision carries the larger, not the smaller, measured skill advantage. A third, independent region corroborates the same decoupling in the other direction: the Baltic Sea's lift over persistence is the largest of the five regions at every horizon tested, while its base rate is the second-lowest. The same precision figure driving this paper's central contrast, 0.202, is also the number a live product decision inside this codebase already used to scope the Arabian Sea's own HAB alert tool to a three-day rather than a seven-day horizon — a defensible call that, read next to this paper's results, is a statement about how rare blooms are in that one basin, not about how far ahead HAB forecasting works in general. We recommend reporting skill lift over a persistence baseline beside precision, always, rather than ranking regions, systems, or papers on precision alone.

Keywords: harmful algal blooms, precision-recall, class prevalence, skill assessment, persistence baseline, reporting methodology, marine machine learning

1 Introduction

An operational harmful-algal-bloom (HAB) early-warning system, and a paper reporting one, both tend to summarise performance the same way: precision at a fixed, policy-relevant recall, because that is the number an agency experiences

directly at the recall it has already decided it needs. A system built for the Arabian Sea and a system built for the Baltic Sea are then read side by side on that number, as if it measured the same thing in both places. This paper reports a controlled demonstration, from five regional models built on one shared pipeline, that it usually does not: at a fixed target recall, precision is dominated by each region's own bloom base rate, not by how much the model has actually learned about that region's blooms.

The mechanism is not subtle once named. Precision at a fixed recall is a function of the classifier's ranking *and* of the prevalence of the positive class, and prevalence alone moves it a long way even when the ranking is held fixed [2]. What makes this worth demonstrating rather than restating is what happens when two regions are, by construction, at almost exactly the same prevalence: the reporting convention predicts they should look like equally hard (or equally solved) problems, and this paper's central result is a pair of regions where that prediction fails in a way that a live product decision inside this same codebase already had to reckon with, without naming it this way. The codebase's own Arabian Sea HAB alert tool is scoped to a three-day horizon rather than seven, because the seven-day operating point's precision is 0.202 and was judged too many false alarms for that region's product to carry — a defensible call, but one that reads, unlabelled, as a statement about how far ahead HAB forecasting works, rather than about how rare blooms are in that one basin at that one horizon.

1.1 Contributions

1. A **controlled five-region comparison** of HAB early-warning precision, base rate, and skill, all produced by one shared pipeline (§3), removing the usual confound that cross-paper precision comparisons carry every model, feature set, and labelling convention as a bundle.
2. A **matched-prevalence pair as the central evidence**: the Arabian Sea and Bay of Bengal sit within half a percentage point of each other in bloom base rate at every horizon tested — 0.073 against 0.078 at t+3, narrowing to 0.076 against 0.078 by t+7 — and report precision within a few hundredths of each other, while differing by a factor of 2.8 in skill lift over a persistence baseline at the longest horizon — the two regions a precision-only reading would call indistinguishable are the two that most clearly are not.
3. A **third, independent region corroborating the same decoupling in the other direction**: the Baltic Sea's lift over persistence is the largest in the five-region table at every horizon, while its base rate is the second-lowest, showing the base-rate/precision link and the base-rate/lift non-link are not an artefact of the one matched pair.
4. An explicit **reporting recipe** (§5): report precision because it is what a user of the system experiences, report lift over persistence beside it because it is what a modeller needs to judge skill, and do not rank regions, systems, or papers on precision alone.

2 Related Work

Why precision at fixed recall is base-rate-sensitive by construction. Davis and Goadrich [2] establish the formal relationship between ROC space and precision–recall (PR) space and show that, unlike an ROC curve, a PR curve's achievable region is shaped by the class balance of the evaluation set: the same ranking ability produces a different precision–recall curve at a different positive-class prevalence. That result is usually invoked to justify preferring PR-AUC over ROC-AUC on an imbalanced problem, which this codebase's own HAB pipeline already does for exactly that reason (§3). This paper uses the same underlying fact for a different purpose: not to choose a metric within one region, but to show what the metric does *across* regions of differing prevalence, where it is compared as if prevalence were held constant when it is not.

An operational HAB early-warning system reporting skill for one region. Anderson et al. [1] report the initial skill assessment of the California Harmful Algae Risk Mapping (C-HARM) system, an operational domoic-acid and *Pseudo-nitzschia* bloom forecast for the California coast, evaluated against its own regional record. Their assessment is a single-region skill report of exactly the kind this paper's argument is about: it is a real, carefully validated number for the region it was built for, and it is not designed to answer, and does not claim to answer, how that region's reported skill would compare to another region's similarly reported skill if the two differ in bloom prevalence. This paper's pipeline uses an internal persistence baseline as the closest available stand-in for that kind of external operational benchmark, exactly as §3 describes, because a directly comparable external cross-region benchmark is not available for the five basins studied here.

The field's own reporting-discipline literature. Zahir et al. [4] review monitoring, forecasting, and early-warning approaches for harmful algal blooms broadly and describe precision, recall, and related classification metrics as the standard way such systems report skill — consistent with, though not itself a study of, the cross-region comparability question this paper isolates. More directly, Zhu et al. [5] propose the EMBRACE checklist for environmental machine-learning studies, requiring an explicit, reported baseline comparison as a condition of a credible performance claim rather than an optional extra. This paper's recommendation in §5 — report a second, skill-sensitive number beside precision rather than precision alone — is a specific, HAB-early-warning instance of that checklist's general baseline-reporting requirement, evidenced here with a concrete case where its absence would have hidden a real, measured difference.

Relation to a companion catalogue. This result previously appeared as one line of a broader eleven-instance catalogue of silent, plausible, signed-direction failures in one operational marine machine-learning platform [3], where it is one of eleven instances rather than the paper's subject. This paper isolates that one instance, re-derives every number directly from the underlying report files rather than

summarising them, and adds the per-horizon base-rate table and the Baltic Sea corroboration that the catalogue paper's brevity did not have room for.

3 Methods

Regions and shared pipeline. Five regions — Arabian Sea, Bay of Bengal, Baltic Sea, Benguela, and the California Current — are each run through the identical HAB early-warning pipeline (`machine_learning/hab_early_warning/`), changing only the geographic bounding box each region's data is fetched and modelled within. Arabian Sea is this codebase's default, unsuffixed region; its report files carry no region suffix, and the other four carry their region's name. Every region shares the same 6-year training window (2016–2021), the same forecast-then-threshold label framing, the same rolling-origin cross-validation and final chronological split, the same LightGBM architecture and calibration procedure, and the same persistence baseline and operating-point definition; only each region's own geography, and the label thresholds and model fitted on that region's own data, differ. The final chronological holdout that every number in §4 is read from holds 577,608 rows for Arabian Sea, 1,090,600 for Bay of Bengal, 248,296 for Baltic Sea, 655,344 for Benguela, and 424,432 for California Current — constant within a region across the three horizons tested, since only a horizon's label and positive count shift, not the holdout's row count. The Arabian Sea run is documented, in the pipeline's own module readme, as processing roughly 3.84M grid-cell-days end to end across 136 features; this figure describes that one region's pipeline scale and is not claimed for the other four, whose feature sets are not byte-identical (§6).

Label: forecast, then threshold, per region. The target is not a direct bloom classifier. A per-cell, per-season chlorophyll threshold is fit at the 90th percentile, on training rows only, and a bloom is declared where the forecast chlorophyll at a given horizon meets or exceeds that cell's and season's own threshold. Fitting per region (rather than pooling all five regions' chlorophyll distributions before choosing a threshold) is deliberate: a fixed absolute chlorophyll cutoff is not physically comparable across basins of different productivity, and pooling would let one region's distribution set another region's labels. This labelling rule is identical in form across all five regions; what differs is the region-specific chlorophyll distribution it is fit against, which is exactly why the resulting base rate is left to fall out of the data rather than fixed in advance, and why the five regions realise base rates from 0.076 to 0.266 at t+7 despite sharing one percentile rule.

Validation: rolling-origin CV with an embargo, then a final chronological holdout. Model selection uses rolling-origin cross-validation with 4 expanding-window folds: each fold trains on everything before a cutoff and tests on the window after it, with an embargo gap equal to the forecast horizon dropped in between, because a training row within that many days of the test window shares its label's own observation period with it. No fold is drawn at random; the codebase's

validation discipline for this problem is rolling-origin only; there is no random k-fold split anywhere in this pipeline. The number reported in §4 for every region comes from a further, final split held out from all cross-validation: the record is divided chronologically into a 70%/15%/15% train/validation/test sequence, the model is fit on the training slice, isotonic calibration is fit on the validation slice only (never on training, where the class-weighted model is close to perfect, nor on the test slice itself), and every number in §4 is read from the test slice, scored by the calibrated model.

Persistence baseline. Every fold and the final holdout also score a persistence baseline — the squashed standardised chlorophyll anomaly at the time of prediction, read forward as the forecast for $t + h$ — because a genuinely autocorrelated field makes "it is blooming now, so it will be blooming at $t + h$" a demanding benchmark, and a model that cannot beat it has not demonstrated skill regardless of its own PR-AUC. **Skill lift over persistence**, the second number this paper reports beside precision, is PR-AUC of the calibrated LightGBM model minus PR-AUC of the persistence baseline, on the identical held-out rows — the same quantity this pipeline's own training code logs internally as `pr_auc_lift_over_persistence`.

Operating point. Precision is read at the threshold, chosen on the same held-out test slice being scored, that comes closest to a target recall of 0.8 — the standard way to report an operating point for a system with one deployment-relevant recall target, and applied identically across all five regions, so it is not a source of the cross-region difference this paper reports even though it is disclosed here as a limitation of any one region's number in isolation (§6). PR-AUC, by contrast, is threshold-free and is what the lift figure above is built from.

4 Results

Table 1 orders the five regions, at every horizon, from lowest to highest base rate, and precision rises with them: from 0.202–0.187 in the two lowest-prevalence basins to 0.844 in the highest-prevalence one at t+7, a spread of roughly 0.844−0.202 driven by a base-rate spread of 0.266−0.076. Read only this broad trend and the reporting convention this paper argues against looks almost defensible: base rate explains most of the visible spread in precision, so precision is close to being a relabelled base rate.

The Arabian Sea / Bay of Bengal pair. These two regions are the closest-matched pair in the table: base rates of 0.073/0.078 at t+3, narrowing to within three thousandths by t+5 and two thousandths by t+7 (0.075/0.078, then 0.076/0.078). At t+3 the two also order the way base rate predicts: Bay of Bengal's very slightly higher base rate pairs with a very slightly higher precision (0.461 against 0.449). At t+5 and t+7 the order *reverses*: Arabian Sea's precision is the higher of the two (0.280 against 0.272; 0.202 against 0.187) despite its base rate remaining at

Table 1. All 5 regions at three forecast horizons: bloom base rate (holdout positive rate), precision at the operating point closest to recall 0.8, and skill lift over the persistence baseline (calibrated LightGBM PR-AUC minus persistence PR-AUC), all read from each region's own `hab_early_warning*_holdout.csv` and `*_summary.json`.

Region	Horizon	Base rate	Precision	Lift over persistence
Arabian Sea	t+3	0.073	0.449	0.149
	t+5	0.075	0.280	0.109
	t+7	0.076	0.202	0.053
Bay of Bengal	t+3	0.078	0.461	0.225
	t+5	0.078	0.272	0.189
	t+7	0.078	0.187	0.150
Baltic Sea	t+3	0.139	0.699	0.408
	t+5	0.138	0.521	0.347
	t+7	0.138	0.416	0.327
Benguela	t+3	0.153	0.806	0.158
	t+5	0.154	0.659	0.133
	t+7	0.154	0.566	0.111
California Current	t+3	0.267	0.950	0.113
	t+5	0.266	0.904	0.100
	t+7	0.266	0.844	0.076

or below Bay of Bengal's at both horizons. Base rate alone therefore does not even fully explain the ordering within this one closely matched pair, let alone the magnitude of the skill difference between them: at t+7, Bay of Bengal's lift over persistence is 0.150 against Arabian Sea's 0.053 — 2.8× the skill, while reporting the *lower* precision. A reader comparing only the precision column would rank Arabian Sea's model as marginally ahead of Bay of Bengal's at t+7; a reader comparing lift over persistence would rank Bay of Bengal's model as decisively ahead. Both readers are looking at numbers computed correctly from the same holdout; they reach opposite conclusions because precision at a near-tied base rate is not measuring the same thing lift is.

Baltic Sea as independent corroboration. The Arabian Sea / Bay of Bengal pair is a matched-prevalence contrast constructed by which two regions happen to sit closest together in base rate; a single pair, however striking, could in principle be a coincidence of these particular two basins. The Baltic Sea's own numbers argue that it is not. Baltic Sea's lift over persistence is the largest in the entire table at every horizon tested (0.408, 0.347, 0.327 at t+3/t+5/t+7), while its base rate (0.138 at t+7) is the second-lowest of the five and its precision is mid-table — lower than Benguela's at every horizon despite Benguela sitting at a very similar base rate (0.154 against 0.138 at t+7) and carrying a markedly smaller lift (0.111 against 0.327). Base rate and precision move together across Table 1 broadly, as the introduction's contribution states; base rate and lift over persistence do not

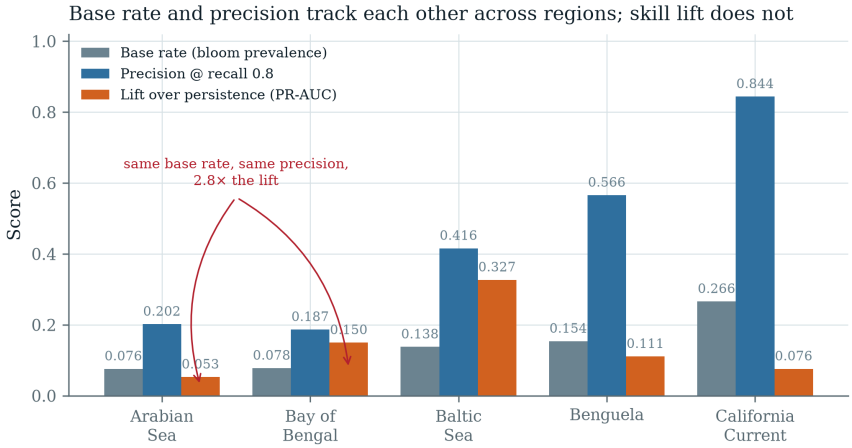


Fig. 1. Base rate, precision at recall 0.8, and lift over persistence, all at t+7, for all five regions ordered by base rate. The first two series move together; the third does not: Arabian Sea and Bay of Bengal are nearly indistinguishable on the first two and differ by 2.8× on the third, and Baltic Sea's lift bar exceeds Benguela's despite a lower base rate.

move together at all, in either the Arabian-Sea/Bay-of-Bengal pair or in Baltic Sea's comparison against Benguela.

5 Discussion

Two readings of Table 1 are both partly right, which is exactly the trap. Read across all five regions, precision at a fixed recall is dominated by base rate — the broad trend is real, base rate is doing most of the visible work, and a system built for a high-prevalence basin will tend to report higher precision than one built for a low-prevalence basin even if the two encode identical skill. Read within the closest-matched pair, precision is not even ordered by base rate at two of the three horizons tested, and the region reporting the lower precision is the one carrying the larger, not the smaller, measured skill advantage. Both facts belong in a report of this kind of system, and the reporting convention this paper argues against keeps only the number that, read alone, suggests the opposite of what the matched pair shows.

The reporting recipe. Quote precision, because it is the number a user of the system actually experiences: an agency deciding whether to act on an alert lives with the false-alarm rate at whatever recall it has chosen to operate at, and that experience does not care why the number is what it is. Quote lift over persistence beside it, because it is the number a modeller needs in order to judge whether the system has learned something a naive, nearly-free baseline had not already captured, and that judgment *does* depend on removing the base-rate confound

precision alone carries. Do not rank regions, systems, or papers on precision alone — not because precision is computed wrong, but because, as Table 1 and [2] both show, it answers a question about the water and the operating point together, not a question about the model in isolation. This is a specific, measured instance of the general baseline-reporting requirement Zhu et al.'s checklist argues for [5]: report the baseline comparison, not only the headline metric, and let a reader see both.

A caveat this codebase had already written, unlabelled. The Arabian Sea HAB alert tool this platform ships is scoped to t+3 rather than t+7 because t+7's precision, 0.202, was judged too many false alarms for a production alert. That is a reasonable, region-specific product decision. Read next to Table 1, though, the same precision number is not evidence that seven-day HAB forecasting is inherently harder than three-day HAB forecasting in general — Arabian Sea's own t+7 lift over persistence, 0.053, is the smallest in the whole table, while Bay of Bengal's t+7 lift, 0.150, is comfortably mid-table despite an even lower t+7 precision. The caveat is correct as a statement about the Arabian Sea specifically and would be overreaching stated as a property of HAB early warning as a method.

6 Limitations

This is a five-region result from one shared pipeline and one label definition, not a survey of independently built HAB early-warning systems, and the claim's external validity is bounded accordingly. The label itself is a weak proxy — forecast chlorophyll exceeding a locally and seasonally fitted 90th-percentile threshold is a stand-in for a bloom, not a verified, in-situ-confirmed one, and says nothing about toxicity — so every number in Table 1 should be read as skill at forecasting this specific weak-label event, not as skill at forecasting harmful blooms in the stronger, agency-relevant sense. The five regions' feature sets are built by the same pipeline logic but are not byte-identical across regions: the Arabian Sea and Bay of Bengal alone carry a monsoon-phase categorical feature that has no equivalent in the other three, since a monsoon phase is not a meaningful seasonal unit in those basins. Every region's reported operating-point precision is read at a threshold chosen on the same held-out slice the precision is then computed on (§3) — standard practice for summarising a deployed operating point, applied identically across all five regions and therefore not itself a source of the cross-region difference reported here, but a reason any single region's precision number, on its own, should not be treated as an unbiased estimate of precision on a future, unseen holdout. Finally, the persistence baseline is this codebase's internal stand-in for an external, cross-system operational benchmark such as C-HARM [1]; no two of the five regions studied here are benchmarked against a shared external system, so the lift figures in Table 1 compare each region's model only to its own region's persistence, not to one another's baselines directly.

7 Conclusion

Five regions, one pipeline, one label definition: precision at a fixed recall tracks each region's own bloom base rate closely enough, read across all five, that the usual cross-region reporting convention looks defensible at a glance. It is not. The Arabian Sea and Bay of Bengal sit within thousandths of each other in base rate, report precision within a few hundredths of each other, and differ by 2.8× in skill lift over a persistence baseline at the longest horizon tested — and at that horizon it is the *lower*-precision region that carries the larger skill advantage, the opposite of what a precision-only reading would conclude. The Baltic Sea corroborates the same decoupling independently, carrying this table's largest lift at the second-lowest base rate. The fix is not a new metric or a new model: it is reporting lift over a persistence baseline beside precision, always, so that a number dominated by which water a system was built for is never mistaken for a number that measures what the system learned about it.

Reproducibility

All five regions are produced by one shared pipeline, `machine_learning/hab_early_warning/`, run once per region with only the geographic bounding box changed (`machine_learning/marine_ml/config.py`'s `HAB_REGIONS`). Every number in Table 1 is read directly from that region's own committed report files, re-verified as of 2026-09-09.

Artifact. Base rate and lift over persistence: each region's `hab_early_warning*_holdout.csv` (`positive_rate` column; PR-AUC of the calibrated LightGBM model minus PR-AUC of the persistence baseline at the matching horizon row — the same quantity `src/train.py` logs as `pr_auc_lift_over_persistence`). Precision: each region's `hab_early_warning*_summary.json`, `"<horizon>" .operating_point.precision`. Holdout size (n): the same `*_holdout.csv` files' `n` column.

Artifact, code. The shared pipeline configuration — percentile threshold, train/validation/test split, fold count, training window, target recall — is read from `machine_learning/hab_early_warning/src/labels.py` (`DEFAULT_PERCENTILE`), `machine_learning/marine_ml/config.py` (`TRAIN_FRACTION`, `VALIDATION_FRACTION`, `HAB_START`, `HAB_END`), `machine_learning/hab_early_warning/src/train.py` (`run()`'s `n_splits=4` default; `target_recall=0.8` passed to `threshold_for_recall` in `fit_final()`), and `machine_learning/marine_ml/validation/splits.py` (`rolling_origin_splits`'s `gap_days` defaulting to the forecast horizon).

Documentation, Arabian Sea only. Pipeline scale (3.84M grid-cell-days, 136 features) is descriptive prose in

`machine_learning/hab_early_warning/readme.md`, about the Arabian Sea run specifically — not re-derived from a report file and not claimed to hold for the other four regions.

References

1. Anderson, C.R., Kudela, R.M., Kahru, M., Chao, Y., Rosenfeld, L.K., Bahr, F.L., Anderson, D.M., Norris, T.A.: Initial skill assessment of the California harmful algae risk mapping (C-HARM) system. Harmful Algae **59**, 1–18 (2016). `https://doi.org/10.1016/j.hal.2016.08.006`
2. Davis, J., Goadrich, M.: The relationship between precision-recall and ROC curves. In: Proceedings of the 23rd International Conference on Machine Learning (ICML '06). pp. 233–240 (2006). `https://doi.org/10.1145/1143844.1143874`
3. Krishna, D.: Silent, plausible, and signed: Eleven measured failure modes in one operational marine machine-learning pipeline (2026), manuscript, MarisAI
4. Zahir, M., Su, Y., Shahzad, M.I., Ayub, G., Rahman, S.U., Ijaz, J.: A review on monitoring, forecasting, and early warning of harmful algal bloom. Aquaculture **593**, 741351 (2024). `https://doi.org/10.1016/j.aquaculture.2024.741351`
5. Zhu, J.J., Boehm, A.B., Ren, Z.J.: Environmental machine learning, baseline reporting, and comprehensive evaluation: The EMBRACE checklist. Environmental Science & Technology **58**(45), 19909–19912 (2024). `https://doi.org/10.1021/acs.est.4c09611`